

ANNE VICTORIA SEFEN

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SUMMARY

Biomedical engineering student and Division I student-athlete drawn to medical device development through both biomedical research and direct patient care. Experience includes building and testing wearable biosensors, developing translational research protocols, and using medical devices in real clinical settings as an EMT and Patient Care Technician. Interested in contributing to the design, testing, and validation of medical technologies that improve patient care and translate effectively from development to clinical use.

SKILLS

- Engineering Design, Prototyping & Experimental Testing
- Biomedical Instrumentation & Laboratory Equipment
- Data Analysis, Statistical Analysis & Technical Documentation
- Hardware Integration, Soldering & Wiring
- MATLAB, CAD, Python, Excel, ImageJ, BioRender

EXPERIENCE

Undergraduate Research Assistant | [Bin Jiang Laboratory](#), QS Institute for Regenerative Engineering *DECEMBER 2025 – PRESENT*

- Investigate mitochondrial transplantation as a therapeutic delivery platform by characterizing the capacity, directionality, retention, and release kinetics of exogenous NAD⁺ uptake in isolated mitochondria across controlled experimental conditions.
- Design and execute quantitative assays using UV-Vis spectrophotometry, JC-1 fluorescence imaging, mitochondrial size-distribution analysis, and Seahorse extracellular flux analysis to evaluate NAD⁺ loading, membrane potential, structural integrity, and OCR.
- Independently develop and optimize experimental protocols from primary literature for mitochondrial NAD⁺ loading and retention; perform quantitative data and image analysis in Excel and ImageJ and create experimental schematics in BioRender.
- Awarded the Wissner-Slivka Research Fellowship; first author on a selected Northwestern Feinberg School of Medicine Research Day abstract, co-author on a second Research Day abstract, and co-author on an accepted TERMIS conference abstract.

Undergraduate Research Assistant | [Rogers Lab](#), Northwestern University | [Project](#) *JANUARY 2025 – DECEMBER 2025*

- Designed, fabricated, and experimentally validated wearable devices for transepidermal water loss (TEWL) collection, optimizing sensor architecture and system performance for noninvasive assessment of epidermal molecular flux and skin-barrier health.
- Built and integrated hardware prototypes using soldering, wiring, and electronic components; conducted benchtop testing and analyzed sensor outputs in Python to characterize performance, repeatability, and data quality.

FDA Intern | Office of New Drug Products (ONDP) *SEPTEMBER 2023 – MARCH 2024*

- Analyzed pharmaceutical datasets and primary literature to support regulatory evaluations of drug safety, biological activity, and exposure risk, synthesizing findings relevant to mechanism-of-action and pharmacodynamic considerations.
- Performed data validation, quality control checks, and detailed scientific documentation to ensure data integrity, traceability, and reproducibility across multidisciplinary regulatory review workflows.

NASA Intern | Goddard Space Flight Center *JUNE 2023 – SEPTEMBER 2023*

- Designed and executed exposure assessment studies to quantify chemical and airborne hazards, generating reproducible datasets relevant to biological risk evaluation and safety thresholds.
- Processed, analyzed, and documented experimental data using standardized protocols, translating findings into clear technical reports and recommendations for multidisciplinary teams.

Clinical & Emergency Healthcare | Wheaton Rescue Squad, Evanston Hospital, Community Health *MARCH 2023 – PRESENT*

- Deliver patient care as a certified EMT, Patient Care Technician, and Pharmacy Technician, routinely operating and interpreting clinical equipment including cardiac monitors, pulse oximeters, automated blood pressure systems, oxygen-delivery devices, glucometers, and other bedside monitoring technologies.
- Lead community blood drives in partnership with local fire departments, collecting 300+ pints of blood. (bethesdablooddrive.org)

EDUCATION

BS Biomedical Engineering and Economics, [Northwestern University](#)

Expected June '28

Minor in Business Institutions GPA: 3.63 | 2025–26 GPA: 3.83

Relevant Coursework: General Biology I–II; Organic Chemistry I–II with Laboratory; Biomechanics; Materials Science; Physics

Student Leadership

- **D1 Big Ten Cheerleader** - Represent Northwestern Athletics compete nationally at NCA Collegiate Nationals.
- **External Relations Secretary**, Northwestern Emergency Medical Organization
- **Industry Chair**, Biomedical Engineering Society - Connect students with company representatives and professional opportunities.
- **NU Docs Premedical Pathway Program**, Northwestern University Feinberg School of Medicine

AS Computer Science, [Montgomery College](#) | GPA: 3.90

MEDICAL DEVICE PROJECTS

- Diaphragmatic Pacemaker Waterproofing Device (Northwestern Medical Makers)
- Cerebral Palsy Assistive Mobility Device (Northwestern Human Centered Design & Engineering)